Lesson 11: Wave After Wave

Unreal Engine (optional advanced track)

VIDEO GAME CREATION

This week: clearing a wave of enemies no longer means the game is over - it means the NEXT wave begins. Clear all 3, and that's when you really win.

Open your own [YourName]Game project from Lesson 10 - don't start a new one.

Tip: no new assets this week - multiple waves reuse your exact existing enemy spawn logic. This completes the Level 2 Complete milestone!

Setup

1. Open your [YourName]Game project from Lesson 10.
2. In the Level Blueprint, add two new Integer variables: Wave, default value 1, and TotalWaves, default value 3.
3. In WBP_HUD's Designer view, add a fourth Text Block: WaveText, positioned near ScoreText/LivesText.

Level 1 - Minimal (MVP)

Clearing a wave's enemies spawns a fresh wave, up to 3 total. Clearing wave 3 triggers the real win. The current wave number is shown on screen.

4. Extract the spawn For Loop out of your ResetGame Custom Event into its own new Custom Event SpawnWave (code below) - just the loop, nothing else.
5. In ResetGame, add Wave = 1 to your resets, and call SpawnWave at the end instead of spawning directly.
6. In WBP_HUD's UpdateHUD, add a Wave parameter and a Set Text call for WaveText.
7. In HandleEnemyDestroyed, revise the win-check into a nested Branch (code below): if this was the last wave, win as before; otherwise, advance the wave and call SpawnWave again.
8. Click Compile, then Play, and play through - clearing wave 1 or 2 should spawn a fresh wave instead of ending the game; clearing wave 3 should show the win message.

```
Custom Event SpawnWave ->
  For Loop (0 to 4) -> Spawn Actor from Class(BP_Enemy) -> Bind Event to
OnEnemyDestroyed / OnEnemyEscaped

Custom Event ResetGame (revised) ->
  For Each (Get All Actors of Class(BP_Enemy)) -> Destroy Actor
  Score = 0
```

```
  Lives = 3
  Wave = 1
  EnemiesRemaining = 5
  GameActive = true
  Call SetStatus(HUDWidget, "")
  Call UpdateHUD(HUDWidget, Score, Lives, Wave)
  Call SpawnWave

Custom Event HandleEnemyDestroyed (revised) ->
  EnemiesRemaining = EnemiesRemaining - 1
  Score = Score + 10
  Call UpdateHUD(HUDWidget, Score, Lives, Wave)
  Branch(EnemiesRemaining <= 0) -> True:
    Branch(Wave >= TotalWaves) -> True:
      GameActive = false
      Call SetStatus(HUDWidget, "You Win! Press R to Restart")
    False:
      Wave = Wave + 1
      EnemiesRemaining = 5
      Call UpdateHUD(HUDWidget, Score, Lives, Wave)
      Call SpawnWave
```

Alternate Path: C++ (code, optional advanced track)

The same nested win-check pattern works in C++ instead of Blueprint nodes.

9. Extract your enemy-spawning loop into a SpawnWave() function, called from both ResetGame() and the else-branch of your kill-handling function's nested if.
10. In the kill-handling function: if (EnemiesRemaining <= 0) { if (Wave >= TotalWaves) { bGameActive = false; } else { Wave++; EnemiesRemaining = 5; SpawnWave(); } }

This mirrors the Blueprint path's nested Branch exactly - same shape, just from code.

Level 2 - Complete (Alpha/Beta)

Wave transitions are seamless and correct every time - no double-spawns, no skipped waves, exactly 3 waves before the real win, and restarting always returns to wave 1.

- Play a full 3-wave game: confirm waves 1 and 2 each spawn a fresh set of enemies with no win message, and only wave 3 triggers the win message.
- Let Lives run out partway through wave 2 and confirm the lose message still works correctly, not a wave transition.
- Press R after a win or loss and confirm Wave resets to 1, along with Score and Lives.

Level 3 - Refined (Ready to Publish) - pick any that interest you

- Visual: briefly show a "Wave 2!" message via SetStatus with a Delay node clearing it shortly after, distinct from the final win/lose message.
- Structure: notice Call SpawnWave now appears in two places (ResetGame, and HandleEnemyDestroyed's False branch) - the actual spawn loop only exists once, in the SpawnWave Custom Event itself.

If Something's Wrong

- Wave 1 immediately shows the win message: confirm the nested Branch checks Wave >= TotalWaves, not just EnemiesRemaining <= 0 alone.

- New waves never spawn: confirm Call SpawnWave is actually wired inside the False case of the nested Branch.
- Restarting leaves Wave at whatever it was: confirm ResetGame explicitly sets Wave = 1.

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